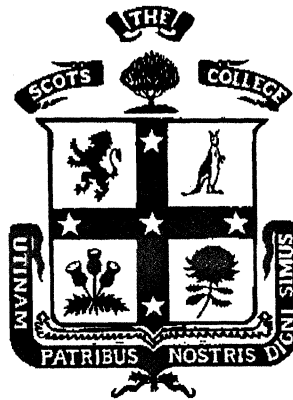


THE SCOTS COLLEGE



YEAR 12

HSC TRIAL EXAMINATION

MATHEMATICS - EXTENSION 1

AUGUST 2005

TIME ALLOWED: 2 HOURS + 5 MINUTES READING TIME

WEIGHTING: 40%

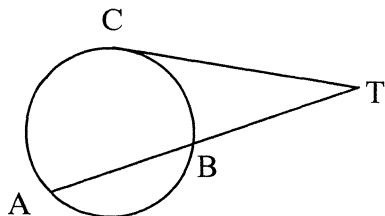
GENERAL INSTRUCTIONS:

- ATTEMPT ALL QUESTIONS.
- USE BLUE/BLACK PEN.
- BOARD APPROVED CALCULATORS MAY BE USED.
- A TABLE OF STANDARD INTEGRALS IS ATTACHED.
- ALL NECESSARY WORKING SHOULD BE SHOWN FOR EACH QUESTION.
- START EACH QUESTION IN A NEW BOOKLET.

a. Solve the inequality $\frac{4-x^2}{x} < 0$

[2]

b.



CT is a tangent to the circle ABC, with ABT a secant intersecting the circle at A and B. Given that $CT = 6$, $BT = 5$ find AB.

[2]

c. The line through the points $A(2, -1)$ and $B(4, 1)$ intersects the line with equation $2y - x + 4 = 0$. Find the acute angle between the lines, to the nearest degree.

[2]

d. It is known that $(x+1)$ is a factor of the polynomial $P(x) = 2x^3 - ax + 2$. Find the value of a .

[1]

e. The line $y = kx$ intersects the circle $x^2 + y^2 - 2x - 14y + 25 = 0$ at two distinct points.

[3]

(i) Show that $25(k^2 + 1) - (7k + 1)^2 < 0$.

(ii) For what value of k is the line $y = kx$ a tangent to the circle?

f. Using the substitution $u = x^2$ find $\int_0^2 x e^{x^2} dx$, leaving your answer in terms of e .

[2]

(a) The function $f(x) = x^3 - \ln(x+1)$ has one root lying between 0.5 and 1. [4]

(i) Show that the root lies between 0.8 and 0.9.

(ii) Use one application of Newton's method to find a second approximation to 3 decimal places if $x = 0.85$ is taken as the first approximation.

(b) [3]

(i) Show that $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x) = 0$ for the domain $0 \leq x \leq 1$.

(ii) Hence, or otherwise, sketch the graph of $y = \sin^{-1} x + \cos^{-1} x$.

(c) A container of hot water at temperature $T^\circ\text{C}$ loses heat when placed in a cooler environment.

It cools according to the rule $\frac{dT}{dt} = k(T - T_0)$, where t is the elapsed time in minutes, T_0 is the environment temperature in degrees ($^\circ\text{C}$). [5]

(i) Show that $T = T_0 + Ce^{kt}$, where C a constant, is a solution to the differential equation.

(ii) A container of water at 90°C is placed in a freezer at -20°C . It cools to a temperature of 60°C in 3 minutes. Find the value of k .

(iii) The same container of water, now at 60°C , is then left in an environment at 20°C . Assuming the value of k remains constant, find, to the nearest degree, the temperature after a further 15 minutes.

- (a) The point $P(3,8)$ divides the interval AB externally in the ratio $k : 1$. If A is the point $(0,2)$ and B is the point $(2,6)$, find the value of k . [3]

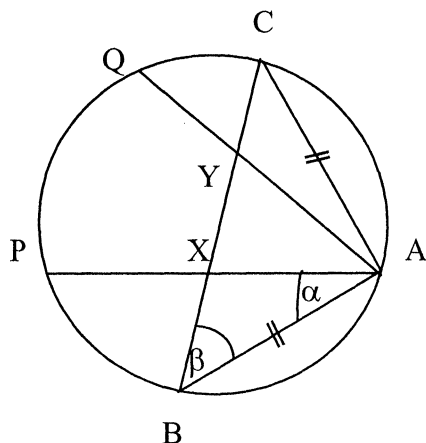
- (b) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. [6]

- (i) Show that the equation of the tangent to the parabola at Q is $y = qx - aq^2$.
- (ii) The tangent at Q and the line through P parallel to the y axis intersect at A . Find the co-ordinates of A .
- (iii) Write down the co-ordinates of M , the midpoint of QA .
- (iv) Determine the locus of M when PQ is a focal chord.

- (c) [3]

- (i) On the same set of axes, sketch the graphs of $y = 2 \cos \theta$ and $y = \frac{1}{2} \theta$ for $-\pi \leq \theta \leq \pi$.
- (ii) Use your sketch to find the number of solutions of the equation $2 \cos \theta = \frac{1}{2} \theta$ for $-\pi \leq \theta \leq \pi$.

(a)



Let ABPQC be a circle with $AB = AC$. Also AP intersects BC at X, and AQ intersects BC at Y. Let $\angle PAB = \alpha$ and $\angle ABC = \beta$. [4]

- (i) Copy the diagram into your Answer Book and state why $\angle AXC = \alpha + \beta$.
- (ii) Show $\angle PQB = \alpha$.
- (iii) Show $\angle AQB = \beta$.
- (iv) Prove $\angle XYQP$ is a cyclic quadrilateral.

(b) Find all angles θ , where $0 \leq \theta \leq 2\pi$, for which $\sqrt{3} \sin 2\theta - \cos 2\theta = 1$. [3]

(c) [2]

- (i) A particle moving in a straight line is subject to an acceleration given by $\ddot{x} = -2e^{-x}$ where x is the displacement from the origin in metres. The particle is initially at the origin with a velocity of 2ms^{-1} . Prove that $v = 2e^{-x/2}$. (You may use the fact that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$).

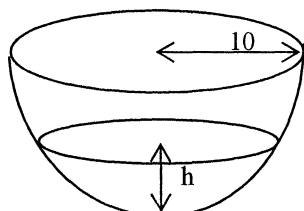
- (ii) Explain the effect on v as x increases without bound.

(d) A particle is moving in simple harmonic motion. It's displacement x at time t is given by $x = 4 \sin(2t + 3)$. [3]

- (i) Find the period of the motion.
- (ii) Find the maximum acceleration of particle.
- (iii) Find the speed of the particle when $x = 2$.

- (a) Prove by mathematical induction that $1 + 3 + 3^2 + 3^3 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$ for all $n \geq 1$. [3]

- (b) [5]



- (i) The diagram above represents a hemispherical bowl of radius 10cm filled with water to a depth of h cm. By finding the volume generated by rotating a circle $x^2 + y^2 = 100$ between $y = 10$ and $y = 10 - h$ about the y -axis show that the volume of water in the bowl is given by $V = \frac{\pi}{3} h^2 (30 - h)$.
- (ii) The hemispherical bowl referred to in (i) above is being filled with water at a constant rate of $2\pi \text{ cm}^3 / \text{min}$. At what rate is the depth increasing when the depth of water is 2cm?

- (c) [4]

- (i) Differentiate $x \cos^{-1} x - \sqrt{1 - x^2}$ with respect to x .
- (ii) Use your result from c (i) above to evaluate $\int_0^1 \cos^{-1} x \, dx$

- (a) [4]
- (i) Solve the equation $2x^3 - 7x^2 - 12x + 45 = 0$, given that two of the roots are equal.
- (ii) The equation in (i) above has two sets of possible solutions. Explain why only one set of values for the roots is valid.
- (b) Find the value of the term that is independent of x in the expansion of $\left(x^2 - \frac{2}{x}\right)^6$. [2]
- (c) Let $f(x) = -\left(\frac{x}{1-x^2}\right)$ [3]
- (i) For what values of x is $f(x)$ undefined?
- (ii) Show that $f'(x) < 0$ at all values of x for which the function is defined.
- (iii) Hence sketch the curve $y = f(x)$.
- (d) [3]
- (i) Show, by means of a sketch, that the curves $y = x^2$ and $y = -\frac{1}{2} \ln x$ meet at a single point.
- (ii) Prove that the tangents to the two curves intersect at right angles at this point of intersection.

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

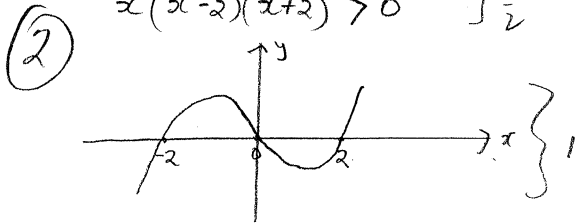
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE : $\ln x \equiv \log_e x, \quad x > 0$

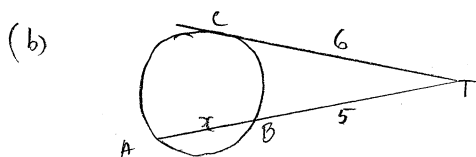
Q1 (12 marks)

(a) $\frac{4-x^2}{x} < 0$
 $\frac{x^2(2-x)(2+x)}{x} < 0 \times x^2 \left. \vphantom{\frac{x^2(2-x)(2+x)}{x}} \right\} \frac{1}{2}$

$x(x-2)(x+2) > 0 \left. \vphantom{x(x-2)(x+2)} \right\} \frac{1}{2}$



$\therefore$ Soln $\{-2 < x < 0\}$ and $\{x > 2\}$



Let $AB = x$

(2) $CT^2 = AT \cdot TB \left. \vphantom{CT^2 = AT \cdot TB} \right\} 1$
 $6^2 = (5+x)5$
 $36 = 25 + 5x$
 $5x = 11$
 $x = 2.2 \text{ cm} \left. \vphantom{5x = 11} \right\}$

(c) Eq. through A, B: $\frac{y-1}{x-4} = \frac{-1-1}{2-4} = \frac{-2}{-2} = 1 \left. \vphantom{\frac{y-1}{x-4} = \frac{-2}{-2} = 1} \right\} \frac{1}{2}$
 $m_1 = 1$

Eq. line $2y - x + 4 = 0 \Rightarrow y = \frac{1}{2}x - 2 \left. \vphantom{y = \frac{1}{2}x - 2} \right\} \frac{1}{2}$
 $\therefore m_2 = \frac{1}{2}$

Angle between lines: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \left. \vphantom{\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|} \right\} 1$
 $= \left| \frac{1 - \frac{1}{2}}{1 + 1(\frac{1}{2})} \right|$
 $= \left| \frac{\frac{1}{2}}{\frac{3}{2}} \right|$
 $= \frac{1}{3}$
 $\therefore \theta = 18^\circ \text{ (nearest degree)}$

(d) Since $(x+1)$ is a factor $P(-1) = 0 \left. \vphantom{P(-1) = 0} \right\} \frac{1}{2}$
 $P(-1) = 2(-1)^3 - a(-1) + 2$
 $= -2 + a + 2$
 $= a$
 $= 0$
 $\therefore a = 0.$

(e) (i) Since the line intersects the circle

$x^2 + (kx)^2 - 2x - 14(kx) + 25 = 0 \left. \vphantom{x^2 + (kx)^2 - 2x - 14(kx) + 25 = 0} \right\} \frac{1}{2}$

$x^2 + k^2 x^2 - 2x - 14kx + 25 = 0$

$(1+k^2)x^2 - (2+14k)x + 25 = 0 \left. \vphantom{(1+k^2)x^2 - (2+14k)x + 25 = 0} \right\} \frac{1}{2}$

Since there are 2 distinct roots $\Delta > 0$
 $\therefore b^2 - 4ac > 0. \left. \vphantom{b^2 - 4ac > 0} \right\} \frac{1}{2}$

$[-2(1+7k)]^2 - 4(1+k^2)25 > 0$

$4(7k+1)^2 - 4(25)(1+k^2) > 0 \left. \vphantom{4(7k+1)^2 - 4(25)(1+k^2) > 0} \right\} \frac{1}{2}$

$(7k+1)^2 - 25(1+k^2) > 0$

ie $25(k^2+1) - (7k+1)^2 < 0$

(ii) The line $y = kx$ is tangential when only one root exists ie $\Delta = 0$

$25(k^2+1) - (7k+1)^2 = 0$

$25k^2 + 25 - 49k^2 - 14k - 1 = 0$

$-24k^2 - 14k + 24 = 0$

$12k^2 + 7k - 12 = 0$

$(4k-3)(3k+4) = 0$

$\therefore k = \frac{3}{4}, -\frac{4}{3}$

(f) Let $u = x^2$ when $x = 0, u = 0$
 $\therefore du = 2x dx$ $x = 2, u = 4 \left. \vphantom{x = 2, u = 4} \right\} 1$

$\int_0^2 x e^{x^2} dx = \frac{1}{2} \int_0^2 2x e^{x^2} dx$

$= \frac{1}{2} \int_0^4 e^u du$

$= \frac{1}{2} [e^u]_0^4$

$= \frac{1}{2} (e^4 - e^0)$

$= \frac{1}{2} (e^4 - 1)$

Q2.

(a) (i) $f(x) = x^3 - \ln(x+1)$

$$f(0.8) = (0.8)^3 - \ln(1.8) \\ = 0.512 - 0.588 \\ < 0$$

$$f(0.9) = (0.9)^3 - \ln(1.9) \\ = 0.729 - 0.642 \\ > 0$$

Since $f(0.8)$ and $f(0.9)$ lie on opposite sides of the x axis, a root lies between 0.8 and 0.9

(ii) $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

(4) $f'(x) = 3x^2 - \frac{1}{x+1}$

$$f'(0.85) = 3(0.85)^2 - \frac{1}{1.85} \\ = 1.627$$

$$x_1 = 0.85 - \frac{-0.001}{1.627} \\ = 0.851$$

$$f(x) = x^3 - \ln(x+1)$$

$$f(0.85) = (0.85)^3 - \ln(1.85) \\ = -0.001$$

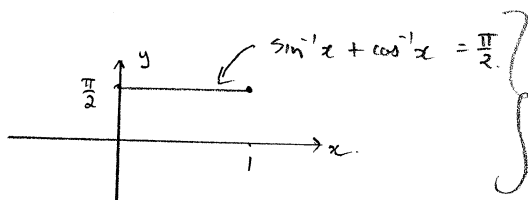
(b) (i) $\frac{d}{dx}(\sin^{-1}x + \cos^{-1}x)$

$$= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \\ = 0$$

(ii) Since $f'(x) = 0$, $f(x) = C$, a constant

When $x = 0$, $f(0) = \sin^{-1}0 + \cos^{-1}0 \\ = \frac{\pi}{2}$

(3) $\therefore f(x) = \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$



(c) (i) $T = T_0 + Ce^{kt}$

$$\frac{dT}{dt} = kCe^{kt} \\ = k(T - T_0)$$

$\therefore T = T_0 + Ce^{kt}$ is a soln of $\frac{dT}{dt} = k(T - T_0)$

(ii) At $t = 0$, $T = 90$
 $t = 3$, $T = 60$

$$T_0 = -20$$

$$T = T_0 + Ce^{kt}$$

$$90 = -20 + Ce^0$$

$$\therefore C = 110$$

$$T = T_0 + 110e^{kt}$$

For $t = 3$, $60 = -20 + 110e^{3k}$

$$80 = 110e^{3k}$$

$$\therefore e^{3k} = \frac{8}{11} \\ 3k = \ln\left(\frac{8}{11}\right)$$

$$k = \frac{1}{3} \ln \frac{8}{11}$$

$$= -0.106 \dots (3 \text{ dp})$$

(ii) At $t = 0$, $T = 60$

$$T_0 = 20$$

$t = 15$, $T = ?$

$$T = T_0 + C_1 e^{kt}$$

$$60 = 20 + C_1 e^0$$

$$\therefore C_1 = 40$$

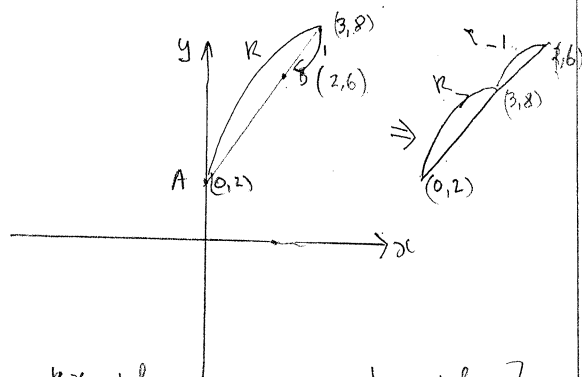
$$T = T_0 + 40e^{kt}$$

$$T = 20 + 40e^{15(-0.106)}$$

$$= 28^\circ \text{C}$$

Q 3.

(a).



$$x_p = \frac{kx_2 + lx_1}{k+l}; \quad y_p = \frac{ky_2 + ly_1}{k+l}$$

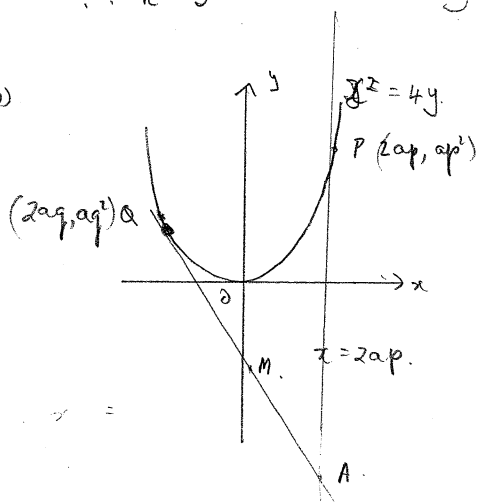
(3)

$$3 = \frac{2k-1(0)}{k-1} \quad \theta = \frac{6k-2(1)}{k-1}$$

$$3k-3 = 2k$$

$$\therefore k = 3$$

(b)



$$\begin{aligned} (1) \quad & x^2 = 4ay \\ & 2x = 4ay' \\ & y' = \frac{2x}{4a} = \frac{2 \cdot 2aq}{4a} = q \end{aligned}$$

$$\text{eq. tangent at Q : } \frac{y-aq^2}{x-2aq} = q$$

$$\begin{aligned} y - aq^2 &= qx - 2aq^2 \\ y &= qx - aq^2 \end{aligned}$$

(6)

(ii) Point of intersection A :

$$y = qx - aq^2 \text{ and } x = 2ap$$

$$\therefore y = q \cdot 2ap - aq^2 = 2apq - aq^2$$

$$\therefore A \text{ has co-ords } [2ap, (2apq - aq^2)]$$

$$\begin{aligned} \text{(iii) Co-ords of M : } & \left(\frac{2ap + 2aq}{2}, \frac{aq^2 + 2apq - aq^2}{2} \right) \\ & = [a(p+q), apq] \end{aligned}$$

(iii) If PQ is a focal chord $pq = -1$.

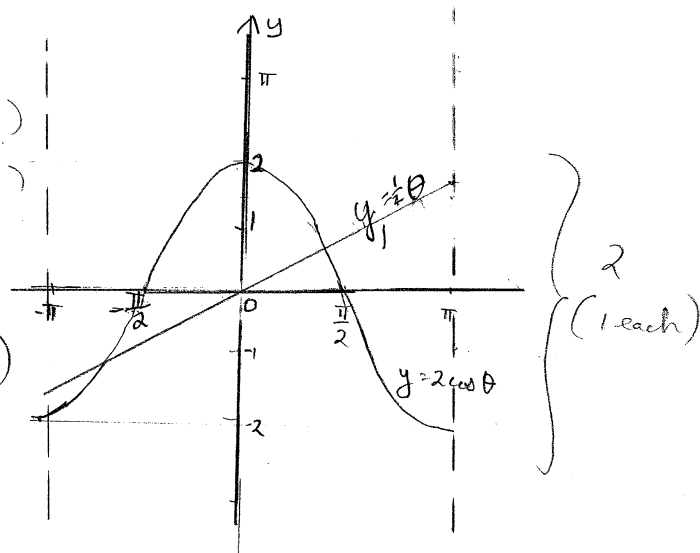
$$\text{Now, } x = a(p+q), y = apq$$

$$\therefore y = a(-1) = -a$$

$$\text{locus of M is } y = -a$$

(c)

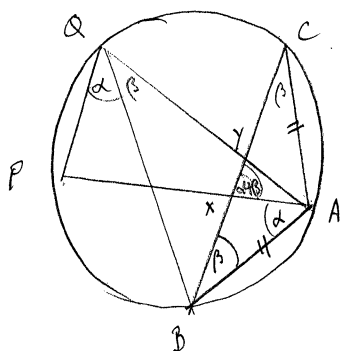
(1)



(3)

$$(ii) \text{ No solutions ; } 2$$

Q4.



(i) $\angle AXC = \alpha + \beta$ (ext. $\angle$ of $\triangle AXB$)

(ii) $\angle PQB = \angle PAB = \alpha$ (angles at circumference on same arc PB)

(4)

(iii) $\angle AQB = \angle BCA$ (angles on same arc AB)
 $= \angle ABC$ (eq. $\angle$'s, isos $\triangle ABC$)
 $= \beta$

(iv) $\angle YXA = \angle YQP = \alpha + \beta$
 $\therefore XYQP$ is cyclic (ext $\angle$ eqs int opp $\angle$)

b) $\sqrt{3} \sin 2\theta - \cos 2\theta = 1$ $\therefore 0 \leq 2\theta \leq 4\pi$
 $\frac{\sqrt{3}}{2} \sin 2\theta - \frac{1}{2} \cos 2\theta = \frac{1}{2}$ $\sin \alpha = \frac{1}{2}$
 $\sin 2\theta \cos \alpha - \cos 2\theta \sin \alpha = \frac{1}{2}$ $\alpha = \frac{\pi}{6}$
 $\sin(2\theta - \alpha) = \frac{1}{2}$

(3) $\therefore 2\theta - \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$
 $2\theta = \frac{\pi}{3}, \pi, \frac{14\pi}{6}, \frac{18\pi}{6}$
 $= \frac{\pi}{3}, \pi, \frac{7\pi}{3}, 3\pi$
 $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}$

(c) (i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -2e^{-x}$
 $\frac{1}{2} v^2 = -2 \int e^{-x} dx$
 $= -2 \frac{e^{-x}}{-1} + C$
 $\frac{1}{2} v^2 = 2e^{-x} + C$ $\left. \vphantom{\frac{d}{dx}} \right\} 1/2$

at $t=0$, $x=0$, $v=2$.
 $\therefore \frac{1}{2} \cdot 4 = 2e^0 + C$
 $2 = 2 + C$
 $\therefore C = 0$
 $\frac{1}{2} v^2 = 2e^{-x}$
 $v^2 = 4e^{-x}$
 $v = \sqrt{4e^{-x}}$
 $= 2e^{-\frac{x}{2}}$ $\left. \vphantom{\frac{d}{dx}} \right\} 1/2$

(ii) as $x \rightarrow \infty$, $v \rightarrow 2e^{-\frac{\infty}{2}} \rightarrow 0$.
as x increases without bound $v \rightarrow 0$

(d) $x = 4 \sin(2t+3)$ is of form $x = a \sin(nt+d)$
(i) Period $T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$ sec.

(ii) $\ddot{x} = -n^2 x$
Max acc when $x = a$.
 $\therefore \ddot{x} = -2^2 \cdot 4 = -16 \text{ m/s}^2$

(3) (iii) $v^2 = n^2(a^2 - x^2)$
 $= 2^2(4^2 - 2^2)$
 $= 4(12) = 48$
 $v = \sqrt{48}$
 $= 4\sqrt{3} \text{ m/s}$

Q5

(a) Assume $1+3+3^2+\dots+3^{n-1} = \frac{3^n-1}{2}$ is true
 When $n=1$, LHS = 1
 RHS = $\frac{3^1-1}{2} = 1$
 $\therefore$ assumption true for $n=1$

Assume $1+3+3^2+\dots+3^{n-1} = \frac{3^n-1}{2}$ is true for $n=k$

$\therefore 1+3+3^2+\dots+3^{k-1} = \frac{3^k-1}{2}$ is true

For $n=k+1$

$$\begin{aligned} 1+3+3^2+\dots+3^{k-1}+3^k &= \frac{3^k-1}{2} + 3^k \\ &= \frac{3^k-1+2\cdot 3^k}{2} \\ &= \frac{3^{k+1}-1}{2} \end{aligned}$$

(3)

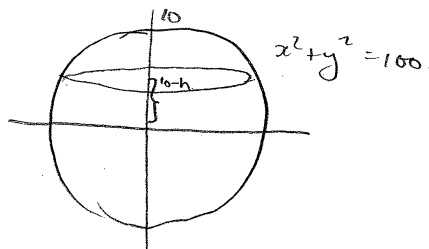
$\therefore$ statement true for $n=k+1$

Since statement is true for $n=1$, it is true for $n=2$

Since statement is true for $n=2$, it is true for $n=3$ and so on.

$\therefore$ statement is true for all $n \geq 1$

(b)



$$\begin{aligned} V &= \pi \int_{10-h}^{10} x^2 dy \\ &= \pi \int_{10-h}^{10} (100-y^2) dy \end{aligned}$$

(5)
$$= \pi \left[100y - \frac{y^3}{3} \right]_{10-h}^{10}$$

$$= \pi \left[\left(1000 - \frac{10^3}{3} \right) - \left(100(10-h) - \frac{(10-h)^3}{3} \right) \right]$$

$$= \pi \left[1000 - 1000 + 100h + \frac{(10-h)^3}{3} - \frac{10^3}{3} \right]$$

$$= \frac{\pi}{3} [300h + (10-h)^3 - 10^3]$$

$$\begin{aligned} &= \frac{\pi}{3} [300h + 1000 - 300h + 30h^2 - h^3 - 1000] \\ &= \frac{\pi}{3} [30h^2 - h^3] \\ &= \frac{\pi}{3} h^2 [30-h] \end{aligned}$$

(11)
$$V = \frac{\pi}{3} (30h^2 - h^3)$$

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$\frac{dV}{dt} = \frac{\pi}{3} (60h - 3h^2) \frac{dh}{dt}$$

$$2\pi = \frac{\pi}{3} [60(2) - 3(2^2)] \frac{dh}{dt}$$

$$6 = (120 - 12) \frac{dh}{dt}$$

$$6 = 108 \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{6}{108} = \frac{1}{18} \text{ cm/sec}$$

$$= 0.55 \text{ mm/sec}$$

(c) (1)
$$\begin{aligned} \frac{d}{dx} (x \cos^{-1} x - \sqrt{1-x^2}) \\ &= \frac{d}{dx} (x \cos^{-1} x - (1-x^2)^{1/2}) \\ &= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} - \frac{1}{2} \frac{(-2x)}{\sqrt{1-x^2}} \end{aligned}$$

(4)
$$\begin{aligned} &= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} \\ &= \cos^{-1} x \end{aligned}$$

(11)
$$\begin{aligned} \cos^{-1} x &= \frac{d}{dx} (x \cos^{-1} x - \sqrt{1-x^2}) \\ \int_0^1 \cos^{-1} x &= \int_0^1 \frac{d}{dx} (x \cos^{-1} x - \sqrt{1-x^2}) \\ &= [x \cos^{-1} x - \sqrt{1-x^2}]_0^1 \\ &= (\cos^{-1} 1 - 0) - (0 - 1) \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

Q6.

(a) (i) let the roots be α, α, β

sum of roots $\alpha + \alpha + \beta = -\frac{b}{a}$

product pairs $\alpha\alpha + \alpha\beta + \alpha\beta = \frac{c}{a}$ $\frac{1}{2}$

product of roots $\alpha\alpha\beta = -\frac{d}{a}$

$2\alpha + \beta = \frac{7}{2}$

$\alpha^2 + 2\alpha\beta = -\frac{12}{2} = -6$

$\alpha^2\beta = -\frac{45}{2}$

From (1) $\beta = \frac{7}{2} - 2\alpha$

From (2) $\alpha^2 + 2\alpha(\frac{7}{2} - 2\alpha) = -6$

$\alpha^2 + 7\alpha - 4\alpha^2 = -6$

$3\alpha^2 - 7\alpha - 6 = 0$

$(3\alpha + 2)(\alpha - 3) = 0$

$\therefore \alpha = 3, -\frac{2}{3}$

$\beta = -\frac{5}{2}, \frac{29}{6}$

$\therefore$ possible roots are $3, 3, -\frac{5}{2}$ and $-\frac{2}{3}, -\frac{2}{3}, \frac{29}{6}$

(ii) Substitution of $-\frac{2}{3}, \frac{2}{3}, \frac{29}{6}$ in $\alpha^2\beta > 0$ not $-\frac{45}{2}$

$\therefore$ only roots are $3, 3, -\frac{5}{2}$

(b) $T_{k+1} = {}^6C_k (x^2)^{6-k} (-2x^{-1})^k$

$= {}^6C_k x^{12-2k} (-2)^k x^{-k}$

$= {}^6C_k (-2)^k x^{12-3k}$

Term independent when $12-3k=0$

$\therefore k=4$

$\therefore T_5 \text{ required} = {}^6C_4 (-2)^4 = 15 \times 16 = 240$

(c) $f(x) = -\left(\frac{x}{1-x^2}\right) = \frac{-x}{(1+x)(1-x)}$

(i) Since $(1-x)(1+x) \neq 0$

(3) $\therefore x$ undefined at $x=1, -1$

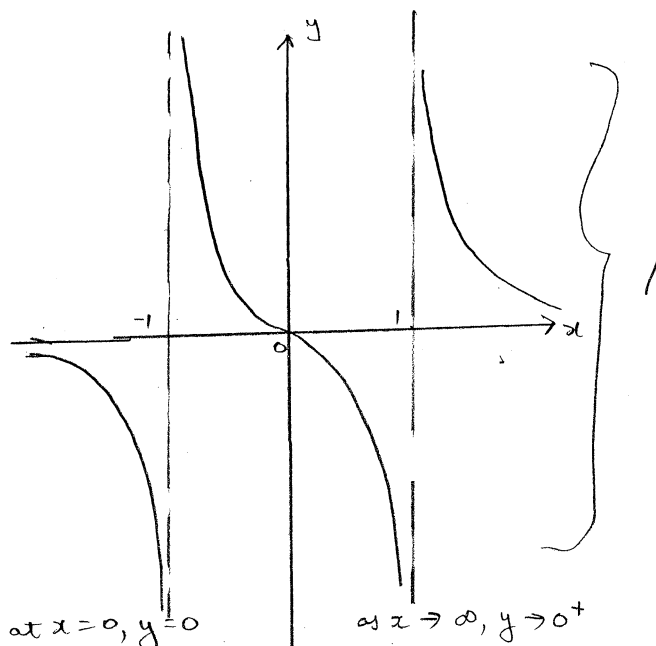
(ii) $f(x) = \frac{-x}{1-x^2}$

$f'(x) = \frac{(1-x^2)(-1) - (-x)(-2x)}{(1-x^2)^2}$

$= \frac{-1+x^2-2x^2}{(1-x^2)^2}$

$= \frac{-(1+x^2)}{(1-x^2)^2}$

< 0 since $1+x^2, (1-x^2)^2 > 0$
for all $x, x \neq -1, 1$



at $x=0, y=0$

as $x \rightarrow 1^+, y \rightarrow \infty$

as $x \rightarrow 1^-, y \rightarrow -\infty$

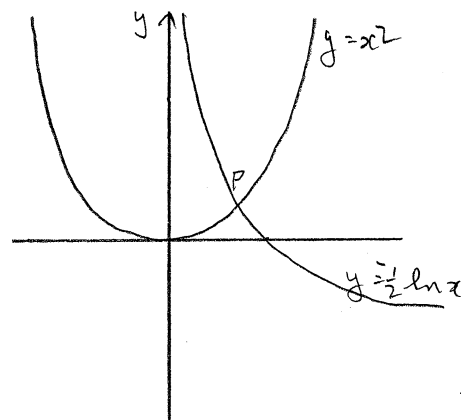
as $x \rightarrow -1^+, y \rightarrow -\infty$

as $x \rightarrow -1^-, y \rightarrow \infty$

as $x \rightarrow \infty, y \rightarrow 0^+$

as $x \rightarrow -\infty, y \rightarrow 0^-$

(d) (i)



(3)

(ii) $y_1 = x^2$

$y_1' = 2x$

$y_2 = \frac{1}{2} \ln x$

$y_2' = \frac{1}{2x}$

let P have abscissa $x=a$

$\therefore m_1 = 2a$

$m_2 = -\frac{1}{2a}$

$m_1 m_2 = 2a \times -\frac{1}{2a} = -1$

$\therefore$ tangents are perp. at P.

Q7.

a 4
b 8

(a)(i)

$$(1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_r x^r + \dots + {}^{2n}C_{2n} x^{2n}$$

$$= 1 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_r x^r + \dots + x^{2n}$$

(ii) When $x=1$

$$2^{2n} = {}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \dots + {}^{2n}C_n + \dots + {}^{2n}C_{2n-1} + {}^{2n}C_{2n}$$

$$= [{}^{2n}C_0 + {}^{2n}C_1 + \dots + {}^{2n}C_n] - {}^{2n}C_n + [{}^{2n}C_n + \dots + {}^{2n}C_{2n}]$$

$$= 2[{}^{2n}C_0 + {}^{2n}C_1 + \dots + {}^{2n}C_n] - {}^{2n}C_n$$

$$\therefore 2[{}^{2n}C_0 + {}^{2n}C_1 + \dots + {}^{2n}C_n] = 2^{2n} - {}^{2n}C_n$$

$$2 \sum_{r=0}^n {}^{2n}C_r = 2^{2n} - {}^{2n}C_n$$

4

$$\sum_{r=0}^n {}^{2n}C_r = 2^{2n-1} - \frac{1}{2} {}^{2n}C_n$$

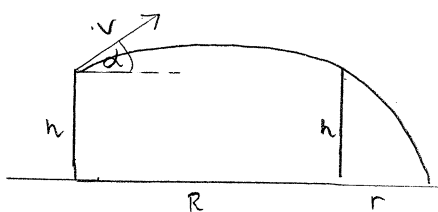
$$= 2^{2n-1} - \frac{1}{2} \frac{(2n)!}{(n)!(n)!}$$

$$= 2^{n-1} - \frac{(2n)!}{2(n!)^2}$$

(b)

(i)

8



Horizontally

$$\ddot{x} = 0$$

$$\therefore \dot{x} = \int 0 dt = C_1$$

$$\text{at } t=0, \dot{x} = v \cos \alpha$$

$$x = \int \dot{x} dt = \int (v \cos \alpha) dt = Vt \cos \alpha + C_2$$

$$\text{at } t=0, x=0 \therefore C_2 = 0$$

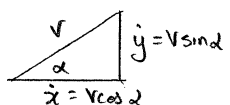
$$x = Vt \cos \alpha$$

Vertically $\ddot{y} = -g$

$$\dot{y} = \int g dt = -gt + C_1$$

$$\text{at } t=0, \dot{y} = v \sin \alpha$$

$$\therefore C_1 = v \sin \alpha$$



$$\dot{y} = -gt + v \sin \alpha$$

$$y = \int \dot{y} dt = \int (-gt + v \sin \alpha) dt$$

$$= -\frac{1}{2} gt^2 + Vt \sin \alpha + C_2$$

$$\text{at } t=0, y=h \therefore h = C_2$$

$$y = -\frac{1}{2} gt^2 + Vt \sin \alpha + h$$

$$(ii) \text{ From (i), } t = \frac{x}{v \cos \alpha}$$

$$y = -\frac{1}{2} g \left(\frac{x}{v \cos \alpha} \right)^2 + v \left(\frac{x}{v \cos \alpha} \right) \sin \alpha + h$$

$$= -\frac{1}{2} g \frac{x^2}{v^2 \cos^2 \alpha} + x \tan \alpha + h$$

$$y = h + x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$$

(iii) The ball clears the fence when $y \geq h, x \geq R$

$$\therefore h + R \tan \alpha - \frac{gR^2}{2v^2 \cos^2 \alpha} \geq h$$

$$R \tan \alpha - \frac{gR^2}{2v^2 \cos^2 \alpha} \geq 0$$

$$\frac{R^2 g}{2v^2 \cos^2 \alpha} \leq R \tan \alpha$$

$$\frac{2v^2 \cos^2 \alpha}{R^2 g} \geq \frac{1}{R \tan \alpha}$$

$$v^2 \geq \frac{R^2 g}{(R \tan \alpha) (2 \cos^2 \alpha)}$$

$$v^2 \geq \frac{gR}{2 \tan \alpha \cos^2 \alpha}$$

$$v^2 \geq \frac{gR}{2 \sin \alpha \cos \alpha}$$

(iv) at C, $y=0, x=R+r$

$$\therefore 0 = h + (R+r) \tan \alpha - \frac{(R+r)^2 g}{2v^2 \cos^2 \alpha}$$

$$-\frac{(R+r)^2 g}{2v^2 \cos^2 \alpha} = (R+r) \tan \alpha + h$$

$$\frac{g}{2 \cos^2 \alpha} = v^2 \left[\frac{\tan \alpha}{(R+r)} + \frac{h}{(R+r)^2} \right]$$

$$\geq \frac{gR}{2 \sin \alpha \cos \alpha} \left[\frac{\tan \alpha}{(R+r)} + \frac{h}{(R+r)^2} \right]$$

$$\frac{1}{\cos \alpha} \geq \frac{R}{\sin \alpha} \left[\frac{\tan \alpha}{(R+r)} + \frac{h}{(R+r)^2} \right]$$

$$\frac{\sin d}{\cos d} \gg \frac{R \tan d}{(R+r)} + \frac{Rr}{(R+r)^2}$$

$$\tan d - \frac{R \tan d}{R+r} \gg \frac{Rr}{(R+r)^2}$$

$$\tan d \left(1 - \frac{R}{R+r}\right) \gg \frac{Rr}{(R+r)^2}$$

$$\tan d \left[\frac{R+r-R}{R+r}\right] \gg \frac{Rr}{(R+r)^2}$$

$$\tan d \left(\frac{r}{R+r}\right) \gg \frac{Rr}{(R+r)^2}$$

$$\tan d \gg \frac{Rr}{r(R+r)}$$

$$(v) \quad V^2 \gg \frac{gR}{2 \cos d \sin d}$$

$$2 \sin d \cos d \gg \frac{gR}{V^2}$$

$$\sin 2d \gg \frac{gR}{V^2}$$

$$\gg \frac{10(80)}{50^2} = \frac{800}{2500} = 0.32$$

$$\therefore 2d \gg 18.66^\circ, 161.34^\circ$$

$$\alpha \gg 9.33^\circ, 80.67^\circ$$

Since closest distance required

$$d = 80.67^\circ$$

$$\tan d \gg \frac{Rr}{r(R+r)}$$

$$\tan 80.67^\circ \gg \frac{80(1)}{r(80+r)}$$

$$r^2 + 80r \gg \frac{80}{\tan 80.67}$$

$$r^2 + 80r - 13.14 \gg 0$$

$$r = \frac{-80 \pm \sqrt{(-80)^2 - 4(1)(-13.14)}}{2}$$

$$= \frac{-80 \pm 80.32}{2}$$

$$= \frac{0.32}{2}, \frac{-160.32}{2}$$

$$= 0.16, -80.16$$

Since $r > 0$, $r = 0.16$ m i.e. 16 cm beyond B